UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 21

Fall 2025

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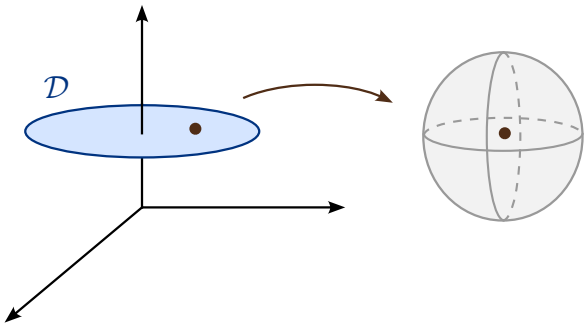
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Recall that to solve $\min f(\boldsymbol{x})$, we solve $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ as a necessary condition. Let's generalize this to convex optimization with constraints.

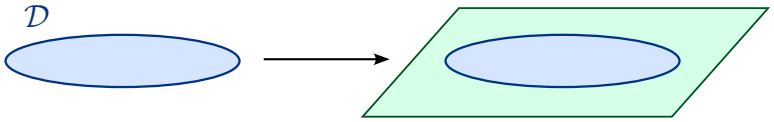
- Given a convex set $\mathcal{D}$, consider its affine hull $\text{aff}(\mathcal{D})$.
- A point $\boldsymbol{x} \in \mathcal{D}$ is said to belong to the relative interior $\text{relint}(\mathcal{D})$ if you can draw a ball in $\text{aff}(\mathcal{D})$ centered at $\boldsymbol{x}$ with non-zero radius that belongs to $\mathcal{D}$.

  Example: consider $\mathcal{D} = \{\boldsymbol{x} \in \mathbb{R}^3 \mid x_3 = 1,\ x_1^2 + x_2^2 \leq 1\}$. There is no interior of $\mathcal{D}$:

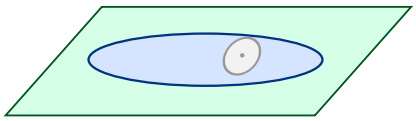


if you draw a ball in $\mathbb{R}^3$ around any point of $\mathcal{D}$, it falls out of $\mathcal{D}$

  But $\text{aff}(\mathcal{D})$ is a 2-dimensional surface:



  And if you draw a ball in 2D, it could be inside $\mathcal{D}$:



  Hence, although $\text{int}(\mathcal{D})$ is empty, $\text{relint}(\mathcal{D}) = \{\boldsymbol{x} \mid x_3 = 1,\ x_1^2 + x_2^2 < 1\}$.

- Consider convex optimization:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, k, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, m. \end{aligned}$$

  Define $\mathcal{D} = \bigcap_{i=0}^{k} \text{dom}\, f_i$.

  (Since h_j's are affine, their domain is $\mathbb{R}^n$ (defined everywhere).)

- It is said that *Slater's condition* holds if there is a point $\boldsymbol{y} \in \text{relint}(\mathcal{D})$ such that

  (i) for all $j \in \{1, \ldots, m\}$, $h_j(\boldsymbol{y}) = 0$;

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(ii) for all $i \in \{1, \ldots, k\}$, $f_i(\boldsymbol{y}) \begin{cases} \leq 0, & \text{if } f_i \text{ is affine,} \\ < 0, & \text{if } f_i \text{ is non-affine, i.e., non-linear.} \end{cases}$

(Note that $\boldsymbol{y}$ is not unique in general.)

Example:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^2} \quad & -\log(x_1 + 1) + \mathrm{e}^{x_2}, \\ \text{s.t.} \quad & x_1^2 + x_2^2 \leq 2, \qquad (1) \\ & x_1 + x_2 \leq 1, \qquad (2) \\ & x_1 + 2x_2 = 2. \qquad (3) \end{aligned}$$

The domain is $\{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1 > -1\}$. Select $\boldsymbol{y} = [0, 1]^{\mathrm{T}}$, then we can verify that the Slater's condition holds:

(3): $0 + 2 \times 1 = 2$. ✓
(2): $0 + 1 \leq 1$. ✓ ($x_1 + x_2$ is affine)
(1): $0^2 + 1^2 < 2$. ✓ ($x_1^2 + x_2^2$ is non-linear/non-affine)

Theorem 1 (Karush-Kuhn-Tucker (KKT) optimality conditions)**.** *For convex optimization, if Slater's condition holds, then $\boldsymbol{x}^*$ is a global minimum iff there are $\boldsymbol{\lambda}^* = [\lambda_1^*, \ldots, \lambda_k^*]^{\mathrm{T}}$ and $\boldsymbol{\mu}^* = [\mu_1^*, \ldots, \mu_m^*]^{\mathrm{T}}$ (called Lagrange multipliers) such that:*

(1) Primal feasibility: $f_i(\boldsymbol{x}^*) \leq 0$, $i = 1, \ldots, k$; $h_j(\boldsymbol{x}^*) = 0$, $j = 1, \ldots, m$.

(2) Dual feasibility: $\lambda_1^*, \lambda_2^*, \ldots, \lambda_k^* \geq 0$.

(3) Complementary slackness: $\lambda_i^* f_i(\boldsymbol{x}^*) = 0$, $i = 1, \ldots, k$.

(4) Stationarity: $\nabla f_0(\boldsymbol{x}^*) + \sum_{i=1}^k \lambda_i^* \nabla f_i(\boldsymbol{x}^*) + \sum_{j=1}^m \mu_j^* \nabla h_j(\boldsymbol{x}^*) = \boldsymbol{0}$.

(We will discuss the names "primal" and "dual" later.)
Special case: $\min f_0(\boldsymbol{x})$.

- There're no constraints, so Slater's condition holds by convention, and there's no $\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*$. The first three conditions (1), (2), (3) no longer exist.
- The KKT conditions reduce to only (4): $\nabla f_0(\boldsymbol{x}^*) = \boldsymbol{0}$. If $f_0(\boldsymbol{x})$ is convex ($\nabla^2 f_0(\boldsymbol{x}) \succeq \boldsymbol{0}$, $\forall \boldsymbol{x}$), then $\boldsymbol{x}^*$ is a global minimum iff $\nabla f_0(\boldsymbol{x}^*) = \boldsymbol{0}$.

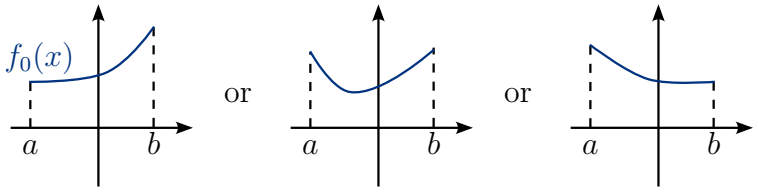
If the optimization is non-convex, then KKT becomes only a necessary but not sufficient condition. KKT in general (non-convex case) is called the first-order necessary optimality condition (FOC).

Example 1 (KKT conditions)**.** Consider

$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f_0(x), \\ \text{s.t.} \quad & a \leq x \leq b, \ a \neq b, \end{aligned}$$

where $f_0(x)$ is convex.

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So, the solution could be on the boundary or interior. Let's find this out using KKT.

$$\begin{aligned} \min \quad & f_0(x), \\ \text{s.t.} \quad & f_1(x) \coloneqq a - x \le 0, \\ & f_2(x) \coloneqq x - b \le 0. \end{aligned}$$

Select $y = (a+b)/2$, then $a - y \le 0$, $y - b \le 0$, so Slater's condition holds.
Define Lagrange multipliers λ_1^* and λ_2^*. KKT:

(1) $a - x^* \le 0$, $x^* - b \le 0$.

(2) $\lambda_1^*, \lambda_2^* \ge 0$.

(3) $\lambda_1^*(a - x^*) = 0$, $\lambda_2^*(x^* - b) = 0$.

(4) $f_0'(x^*) + \lambda_1^*(-1) + \lambda_2^*(+1) = 0$.

There are 7 conditions in (1)–(4). To solve them, we usually start from dual feasibility:

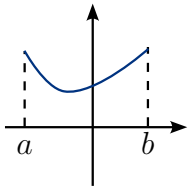
(2) $\lambda_1^* \ge 0 \Rightarrow \lambda_1^* = 0$ or $\lambda_1^* > 0$; $\lambda_2^* \ge 0 \Rightarrow \lambda_2^* = 0$ or $\lambda_2^* > 0$.

There are four scenarios:

(i) $\lambda_1^* = \lambda_2^* = 0$. By stationarity (4),

$$f'(x^*) + 0(-1) + 0(+1) = 0 \;\Rightarrow\; f'(x^*) = 0.$$

If there exists $x^* \in [a, b]$ such that $f'(x^*) = 0$, it is a minimum.

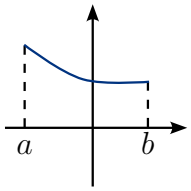


(ii) $\lambda_1^* = 0$, $\lambda_2^* > 0$. By (3), we have $x^* - b = 0 \Rightarrow x^* = b$. By (4), we have

$$f'(x^*) + 0(-1) + \lambda_2^*(+1) = 0 \;\Rightarrow\; 0 < \lambda_2^* = -f'(x^*) = -f'(b).$$

If $f'(b) < 0$, then $x^* = b$ is a minimum.

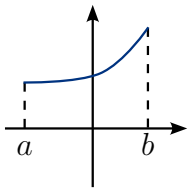
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(iii) $\lambda_1^* > 0$, $\lambda_2^* = 0$. By (3), we have $a - x^* = 0 \Rightarrow x^* = a$. By (4), we have

$$f'(x^*) + \lambda_1^*(-1) + 0(+1) = 0 \;\Rightarrow\; 0 < \lambda_1^* = f'(x^*) = f'(a).$$

If $f'(a) > 0$, then $x^* = a$ is a minimum.



(iv) $\lambda_1^*, \lambda_2^* > 0$. By (3), we have $a - x^* = x^* - b = 0 \Rightarrow a = b$. This is impossible since we assumed $a \neq b$.

In general, for the dual feasibility condition $\lambda_i^* \geq 0,\ i = 1, \ldots, m$, there're 2^m possibilities to find which λ_i^*'s are zero and which are strictly positive. This approach is not efficient and there are numerical algorithms to solve (1), (2), (3), (4) more efficiently.

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